

Task 3 - One-Page Written Summary

Option A: Working Prototype

This Task 3 submission follows Option A: Working Prototype. Building on Task 1, the selected context is a cross-border bank such as HSBC operating across the United States and the European Union. Building on Task 2, the design reflects RegSphere AI Governance Solutions' values of fairness, transparency, accountability, regulatory adaptability, and responsible automation. The prototype is therefore not a replacement for lending officers, lawyers, or compliance professionals. It is a governance and monitoring layer around an AI-supported credit scoring model.

The tool addresses a practical RegTech problem that a static memo or spreadsheet cannot solve well: AI credit models change over time, applicant populations drift, fairness outcomes can deteriorate, and regulatory obligations differ by jurisdiction. The notebook prototype, Task3_demo.ipynb, generates synthetic loan data, trains a simple logistic regression credit scoring model, reports accuracy, AUC, and a confusion matrix, calculates approval disparity, false positive rate gaps, and false negative rate gaps, produces feature importance and rejection reasons, simulates a six-month monitoring period, detects model drift using Population Stability Index, and exports compliance outputs for review.

The jurisdiction-aware logic is based on real regulatory differences. In the United States, the prototype reflects fair lending and adverse action explanation expectations under ECOA, Regulation B, and CFPB guidance on credit decisions using complex algorithms. In the European Union, it reflects the EU AI Act, Regulation (EU) 2024/1689, under which AI systems used to evaluate the creditworthiness of natural persons or establish credit scores are treated as high-risk AI systems, subject to limited exceptions. The EU configuration therefore emphasizes high-risk AI governance, human oversight, documentation, transparency, and post-deployment monitoring.

The core design choice is the jurisdiction rule engine. The same model metrics can lead to different governance results depending on whether the US or EU configuration is active. For example, fairness gaps can trigger a US fair lending review and validation of adverse action explanations, while the EU configuration can additionally require human oversight escalation, risk-management updates, monitoring logs, and technical documentation. The compliance output is exported in jurisdiction_compliance_report.csv, while model monitoring results are exported in monitoring_results.csv.

The prototype also includes a model card and governance documentation stub in model_card_governance_stub.md. This addresses what the tool does, what data it uses, what assumptions it makes, where it can fail, and where human review is required. The numeric thresholds used in the notebook are internal governance thresholds for demonstration. They are not statutory thresholds and should not be read as final legal interpretations.

The tool deliberately does not make final credit approval decisions, prove legal discrimination, issue final adverse action notices, or replace legal and compliance judgement. Synthetic data is used because real cross-jurisdiction consumer lending data is sensitive and limited. The boundary choice is intentional: the prototype demonstrates how a RegTech system can convert real US/EU regulatory divergence into operational monitoring workflows while preserving human accountability for final decisions.